

Financial Text Sentiment Analysis for Markets with Explainable Artificial Intelligence
Project ID# 60

Authors: Yitian Sun, Sirui Li, Jiehao Xing, Ke Zhang, Jalen Xing

Advisors: Adheesh Shenoy, Chan Yang, SheltonAI

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1. Executive Summary

The financial sector generates extensive textual data, yet its utility is often constrained by a lack of explainability [8]. Our work transforms labeled data into explainable datasets, leveraging large language models (LLMs) and BERT (Bidirectional Encoder Representations from Transformers) to generate reasoning for predicted classes. Our methodology provides data pipelines that are critical to SheltonAI, an AI solutions company focused on transparency and control for investors. By analyzing financial text for sentiment and stock-related outcomes for stakeholders, our approach enables stakeholders to uncover meaningful correlations between sentiment values and key financial metrics.

The data pipeline is used to ultimately create a reasoning dataset when inputting financial text and news, which trains models to provide justifications for sentiments. Therefore, sentiment along with reasoning increases trust and usability for decisions involving investments and risks, and allows stakeholders to trace model outputs back to text-based reasoning. Through our experiments, we explore various ways of improving the accuracy of generating explainable sentiment and finding the drivers. Our results conclude that zero-shot and few-shot prompting, along with chain of thought prompting of large language models, improve sentiment prediction accuracy scores from the baseline BERT model and using top-k attention word. The future goals of this project are to expand the dataset and continue supervised fine-tuning on the generated reasoning data, with the aim of building high-accuracy models for financial sentiment analysis with explainable reasoning.

2. Problem Space

2.1 Creating High-Quality Explainable Sentiment for Financial Text

The process of extracting and understanding sentiments from financial texts is crucial in a variety of usage scenarios, including but not limited to trading, market analysis, and investment strategy. The sentiment values in financial articles can significantly impact the market trend, often reflecting and influencing investors' trading behaviors [16]. Although research and techniques have advanced rapidly in the field of natural language processing (NLP) and sentiment analysis in recent years, there are still limitations stemming from the explainability of language models. For instance, it is difficult to accurately reflect how specific texts drive certain sentiment classifications in certain contexts [14]. Nowadays, trending NLP models that perform sentiment analysis mainly rely on black-box approaches, which lack transparency in the face of text input. High-quality explainability is vital in winning the trust among investors and analysts who highly depend on this information to draw insights.

To achieve the goal of providing an explainable sentiment analysis solution, the model should be capable of highlighting specific words or phrases that are responsible for the sentiment

classification decision. Current research has demonstrated that through attention mechanisms or visualization techniques, models can partially reveal their logic behind the sentiment classification process. However, most research often remains superficial or lacks transparency [15]. As a result, developing a comprehensive, easily comprehended method that provides explanations for sentiment classification becomes a pressing challenge. Placing this task within financial scenarios becomes especially demanding due to complex jargon and nuanced expressions.

2.2 Identifying Gaps in Research Between News Sentiment and Market Performance

Although the relationships between sentiments drawn from financial articles and market performance have been a heated topic for decades, significant gaps still remain. Most of the existing literary reviews primarily focus on the short-term impact of sentiment values on the financial market, and they often aggregate sentiment values without adequate differentiation among sentiment types or the underlying drivers of sentiment shifts [13].

Another gap that is noteworthy to address is the separation of sentiments' ability in predicting and understanding market performance across different industry sectors. The relationship between sentiment and market performance varies significantly across industries [15]. For example, a positive correlation between certain financial metrics and sentiment could denote opposite meanings in the consumer discretionary and real estate industries. This is attributed to specific macroeconomic factors, regulatory environments, and industry-specific news cycles. Existing research often neglects these contextual variances, leading to overgeneralization and ineffective investing strategies in certain industries.

To mend the two aforementioned gaps, we will take deeper dives into the impact of different financial metrics across various industries and economic periods. Our project aims to optimize explainability in sentiment analysis models and establish a nuanced understanding of the impact of sentiment values in market performance assessments.

Additionally, our project aims to target three main audiences: financial institutions and banks, fintech startups, and market researchers. People who work at financial institutions and banks can use our pipeline to gather market sentiment to improve and optimize their investment strategies. As for the fintech startups, aside from using our model to understand the underlying market dynamics, they can also integrate our sentiment pipeline to their own financial analysis solutions to quickly adapt to the changing market landscape. Lastly, the market researchers and academics can gather sentiment data using our pipeline to enrich their dataset during model training and testing phase.

3 Existing Solutions

A range of approaches—from lexicon-based methods to deep learning and pre-trained language models—have been developed to analyze sentiment in financial texts. However, these methods have limitations in key areas such as contextual understanding, interpretability, and alignment with real-world financial outcomes.

3.1 Lexicon-Based Approaches

Lexicon-based methods identify sentiment by checking whether words in the text appear in predefined lists of positive or negative terms. These lists, or lexicons, are often tailored to specific domains. In financial contexts, common examples include SentiEcon [31]. These methods do not rely on labeled data, which makes them easy to use in many cases. However, they have several limitations:

- Limited vocabulary: They may not cover new or less common financial terms.
- Lack of context awareness: They cannot interpret meaning based on word combinations or sentence structure. For instance, phrases like “profit down” may be misclassified.
- Weak performance on informal text: They often do poorly on data from sources like social media.

3.2 Machine Learning Approaches

Traditional machine learning models such as Support Vector Machines (SVM) and Random Forest rely on manually engineered features [34][35]. These features often include n-grams, sentiment lexicon scores, and domain-specific indicators. In many cases, these models perform better than lexicon-based methods, but they have several limitations:

- Limited contextual understanding: These models struggle when sentiment is expressed indirectly or through metaphor.
- Dependence on feature quality: They cannot capture deeper semantic meaning when the input features are insufficient.

3.3 Deep Learning Approaches

Neural models such as Recurrent Neural Networks (RNN) and Bidirectional Long Short-Term Memory Networks (Bi-LSTMs) can automatically learn complex semantic features from data [38][39]. These models reduce the need for manual feature engineering and often achieve strong performance in sentiment classification tasks. However, they also present key challenges:

- Risk of overfitting: Performance may degrade on small or imbalanced datasets.
- Lack of interpretability: These models function as black boxes, which limits their use in financial settings where transparency is essential.
- Limited ability to handle long texts: Traditional neural models struggle to retain and utilize information across longer input sequences.

3.4 Pre-trained Language Models

Domain-specific transformer models, such as FinBERT and FinSenticNet, are built by fine-tuning large pre-trained language models on financial texts [36][37]. These models achieve strong performance in sentiment analysis and other financial NLP tasks by leveraging contextual understanding at scale, but they also present important limitations:

- Limited explainability: Interpretations based on attention weights do not provide human-understandable reasoning.
- Shallow pattern recognition: These models often rely on surface-level associations and lack the ability to capture causal relationships between financial events and sentiment.

In conclusion, despite recent advancements, existing sentiment analysis approaches continue to face critical limitations. Most models lack explainability and are unable to produce human-readable justifications for their predictions, which hinders transparency and trust. Contextual understanding remains limited, with models frequently misinterpreting sarcasm, metaphors, and domain-specific expressions common in financial language. Moreover, there is often a disconnect between semantic sentiment and actual market movements, reducing the practical value of these models in real-world financial applications. Finally, current methods offer limited reasoning capabilities, as few models can articulate the underlying rationale behind a sentiment label. This gap further restricts their adoption in high-stakes decision-making environments where interpretability and causal reasoning are essential.

4. High-Level Approach

4.1 Data Curation

The first stage of our project involves a data curation stage to ensure quality, consistency, and representativeness for model training. We collect 6 financial datasets including news articles, financial headlines, and market commentaries from open sources like Kaggle and UCI repository. In addition, we address missing values, formatting errors, and sentiment labeling inconsistencies in these raw dataset. To further enhance dataset quality, we implement data augmentation strategies, including back translation and synonym substitution. These methods significantly enrich our data variety, fix class imbalance issues, and ensure the dataset is available for further training.

4.2 Using the FinBERT Model to Provide Attention Weights

Following data curation, we utilize FinBERT, a specialized variant of the Bidirectional Encoder Representations from Transformers (BERT) model, specifically fine-tuned for financial contexts. The primary reason for selecting FinBERT is its capability to capture nuanced financial language

better than general-purpose language models. An important advantage of FinBERT is its ability to generate attention weights, which indicate the importance of each word in determining sentiment predictions. These attention weights provide valuable insights into the decision-making process of the model, highlighting key words and phrases that significantly influence the sentiment classification outcome.

4.3 Large Language Model Reasoning

In the final step of our high-level approach, we leverage the interpretability provided by FinBERT's attention mechanism by extracting the top-k words with the highest attention scores. These critical words are then integrated into a large language model (LLM) chain-of-thought reasoning framework. Chain-of-thought allows large language models to output reasoning steps [31]. This step aims to enhance transparency by allowing the model to explicitly articulate why specific sentiment classifications were made, thus providing human-readable justifications. Such an approach significantly improves the interpretability and trustworthiness of sentiment predictions, enabling financial analysts and stakeholders to clearly trace sentiment outcomes back to their textual foundations.

4.4 Correlation Analysis Between Market Sentiment & Financial Metrics

We input the collected financial texts into a Fine-Tuned BERT Model and a classifier to generate sentiment scores, and we integrate the companies' normalized financial metrics data. Pearson correlation, multiple linear regression, and time series analysis are then used to calculate the correlation between financial metrics and market sentiment.

5. Data Curation

5.1 Data Collection

Our first step is to collect data for our later model training. We want the training data to be as close to the real financial language as possible. We collate and collect a variety of data sets covering news reports, earnings summaries, financial headlines and market commentary. These contents basically cover the main sources of information that investors are exposed to in their daily lives.

In terms of data collection, we used two methods. On the one hand, we use methods like NewsAPI, as well as some basic web crawlers (such as BeautifulSoup and Selenium) to get the latest financial text. On the other hand, because it was too inefficient for us to label the data manually, we prioritized financial datasets that were open source and tagged with sentiment. We reviewed more than 50 datasets in total, and finally selected several excellent ones in terms of annotation quality and language diversity as the basic data for model training.

Here are parts of examples of our datasets:

- **Dataset 1: Short Financial Sentences**
This dataset contains 211 short financial headlines, with a relatively balanced sentiment distribution: 62 positive, 65 neutral, and 84 negative samples [27].
- **Dataset 2: Full Financial Paragraphs**
This dataset includes 400 paragraph-level financial news articles, offering rich contextual information. The sentiment distribution consists of 215 positive, 1 neutral, and 184 negative samples [28].
- **Dataset 3: Retail Investor Headlines**
This dataset comprises 4,846 financial headlines sourced from sentiment studies focused on retail investors. The sentiment labels include 1,363 positive, 2,879 neutral, and 604 negative samples [29].
- **Dataset 4: Entity-Level Annotated Headlines**
This is the largest dataset in our corpus, containing 10,753 headlines with sentiment annotations at the entity level (e.g., “Infosys”: neutral). The sentiment distribution is relatively balanced: 4,170 positive, 3,444 neutral, and 3,139 negative samples [30].

5.2 Data Preprocessing

Due to the heterogeneous and unstructured nature of our datasets, the next step is data preprocessing to ensure data consistency, integrity and availability. We started by combining all the selected data sources, including Financial PhraseBank, FiQA, and financial sentiment analysis datasets on multiple Kaggle datasets, into a single, unified dataset. Due to the large differences in the structure, field naming, and annotation of these data sets, we adopted a series of standardized processing operations to achieve a uniform format of data.

The first step is data cleansing. We removed all rows containing missing or null values to avoid introducing bias or noise in subsequent model training with incomplete data. At the same time, the encoding format of some text fields is repaired, invalid symbols are removed and Spaces are standardized to ensure the syntactic consistency of the whole data set. Next, we give uniform names to the fields representing the financial text (such as "sentence", "headline", "text") in different datasets, consolidating all the text into a single column named "news". Whether it is a phrase, title, or complete paragraph, all are grouped into this column to ensure the universality of the processing flow and reduce the complexity of subsequent model processing.

In terms of the standardization of emotional labels, we have uniformly converted labels used in different sources (such as "positive", "neutral", "negative" in string form, or 0, 1, 2 in numerical form) into standard numerical codes:

- -1 for negative sentiment
- 0 for neutral sentiment
- 1 for positive sentiment

This numerical processing is convenient for the subsequent training and evaluation of machine learning models, and also avoids the problem of error analysis caused by inconsistent label formats. Finally, we divided the processed dataset into a training set and testing set according to the ratio of 70%:30%. In this process, we ensure that the proportion of emotion labels is consistent among the categories, so as to make the model evaluation results more representative and fair.

5.3 Data Augmentation

After data preprocessing, we performed a statistical analysis of the emotion distribution of the dataset and found that there was a serious category imbalance: about 67% of the samples were positive emotions, 25% were neutral emotions, and only 8% were negative emotions. This imbalance will directly affect the training process of the model, making the model tend to predict the category with high frequency during learning, and the recognition effect of neutral and negative text will be greatly reduced, affecting the overall generalization ability and application credibility.

To solve this problem, we propose a set of hybrid data enhancement strategies specifically designed to expand the amount of data in the neutral and negative categories to align them with the main categories. We have chosen two classical and complementary methods of natural language enhancement: Back Translation and Synonym Substitution.

Method 1: Back Translation

Back Translation is a common data enhancement technique, the core idea of which is to translate a text into another language and then back to the original language, so as to generate new sentences with semantic equivalence but different expression forms. This method can not only effectively improve the linguistic diversity of corpus, but also keep the original emotion label unchanged to the greatest extent, so it is especially suitable for minority class enhancement in sentiment analysis scenarios.

In this project, we used the Google Translate API to translate neutral and negative samples from English to French and then back to English. For example, The original sentence "The stock dropped due to disappointing earnings" may be translated back to "The share price declined

following underwhelming earnings. results ". Although the word order and wording changed, the emotional expression remained consistent, which enhanced the reliability of the samples [9].

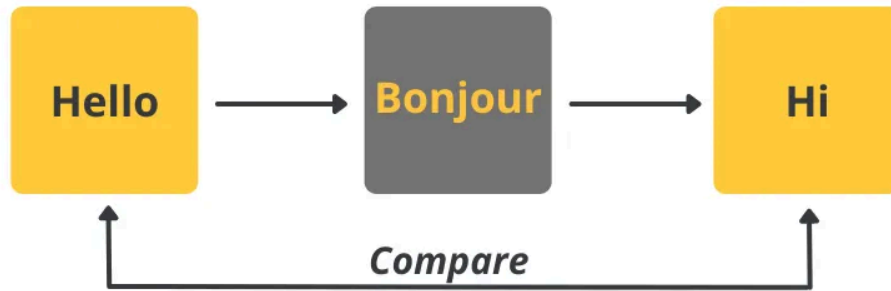


Fig. 1. Back-translation with sampling for low-resource languages. Source: [11].

Method 2: Synonym Substitution

Synonym substitution is a kind of text enhancement method based on lexical level. The main process of synonym substitution is to identify the key words in the sentence and replace them with semantically similar words, so as to generate new samples with diverse language styles and consistent emotions. This method has the advantages of strong controllability and high generation efficiency, which is suitable for constructing a large number of slight variants and enhancing the robustness of the model to lexical diversity. For example, the sentence “The company reported strong earnings this quarter” can be augmented as “The firm announced robust profits this quarter.” In this case, “company” is replaced with “firm,” “reported” with “announced,” and “strong” with “robust.” These substitutions preserve the original positive sentiment but provide the model with new lexical patterns and phrasing structures [10].

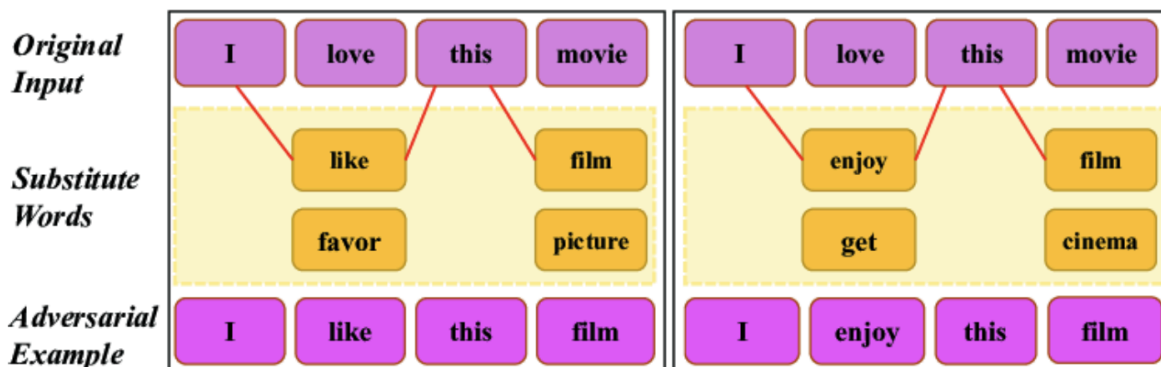


Fig. 2. Examples of sememe-based and synonym-based word substitution. Source [12].

In this project, we use the WordNet in the NLTK toolkit to do this. We first divide each text with neutral or negative emotions, and remove the stop words in it to retain the keywords with actual semantics. Then, randomly select one or two replaceable words from the remaining non-stopping words, look for their synsets in the WordNet thesaurus, and randomly select a word to replace. In the whole process of substitution, the grammatical structure of the sentence should be ensured to avoid obvious speech disorders caused by mismatching of parts of speech or collocation errors. Finally, we ensure enhanced sample quality and emotional consistency by setting rules to screen the replaced sentences and filter out those with significant semantic changes or significant loss of readability.

Method 3: Hybrid enhancement strategy

To further increase the diversity of the enhancement effects, we use a combination of the above two methods: neutral and negative samples are randomly grouped by 50%, one group applies back translation, and the other group applies synonym substitution. This hybrid approach not only introduces the diversity of syntactic structure but also complements the change of lexical level, avoiding overfitting or semantic drift caused by a single enhancement approach.

Finally, we ensured that the whole sample size of each category was roughly equal within the training set (about 33.3% for each category), thus achieving a balance between the sentiment categories in the input data. The enhancement strategy effectively alleviates the class imbalance problem, and provides more stable data support for the fairness evaluation, stability training and generalization performance of the model.

5.4 Data Filtering & Data Quality Classifier

High-quality data is foundational to the success of any sentiment analysis model, especially when the objective is to provide explainable and actionable insights in financial contexts. Inspired by the findings in *Textbooks Are All You Need*, we adopt a similar methodology that emphasizes the importance of clean, well-curated data in enhancing model performance [17]. Their work showcased how textbook-level datasets can significantly improve results by employing rigorous data filtering and decontamination procedures. Building on their foundation, we implemented a classifier-based filtering process to eliminate irrelevant or noisy sentiment data. This step ensures our dataset to only include data that are deemed to be “high quality” by our data quality classifier, which will be further explained below.

Additionally, we applied decontamination techniques such as n-gram overlap analysis and embedding similarity checks. These steps are essential to prevent redundancy, reduce overfitting, and maintain a strict boundary between training and test datasets.

To identify high-quality sentiment datasets, our team incorporated self-consistency using the Phi-2 model, which performs particularly well for language understanding [18]. Our

methodology for data curation was using a high temperature and allowing Phi-2 to provide 5 classifications, determining whether the data was high quality or not, and then use majority voting to filter out the data. This data curation process ensures the reliability and diversity of the data and lays a solid foundation for training our model.

6. Model Architecture & Framework

6.1 FinBERT Fine-Tuning

BERT (Bidirectional Encoder Representations from Transformers) is a widely used deep learning model that captures the contextual meaning of words by processing text bidirectionally (Devlin et al., 2019). It is pretrained on large-scale general-domain corpora such as Wikipedia and BooksCorpus, allowing it to generate robust language representations for a variety of natural language processing (NLP) tasks. However, its effectiveness often diminishes in specialized domains like finance, where the language includes unique terminology and subtle contextual cues.

To establish a performance baseline for financial sentiment analysis, we fine-tuned BERT on domain-specific data, assessing its ability to handle the linguistic characteristics of financial texts prior to applying more targeted adaptations.

Financial sentiment analysis poses distinct challenges that generic language models often fail to address. The language used in financial communication is intricate, featuring technical jargon, implicit sentiment, and highly contextual phrasing. To address this gap, FinBERT—a domain-adapted version of BERT—was developed through continued pretraining on financial news and corporate reports, resulting in significantly improved sentiment classification performance in finance-specific tasks (Araci, 2019).

We further fine-tuned FinBERT using extensive hyperparameter tuning, carefully adjusting batch size, learning rate, and weight decay to ensure the model generalized well across varied financial datasets (Yang et al., 2020).

6.2 Explainability Methods

Explainability is essential in financial applications, where model transparency directly impacts trust and decision-making (Doshi-Velez & Kim, 2017). To enhance interpretability, we applied three complementary methods: attention scores, LIME, and SHAP.

We extracted accumulated attention scores from the transformer layers to identify which tokens the model focused on during prediction. These scores offer insight into the linguistic patterns driving the sentiment classification.

Additionally, we used LIME (Local Interpretable Model-agnostic Explanations) to generate local approximations of the model’s behavior. LIME perturbs input text and fits interpretable models to highlight influential words for individual predictions.

Also, we applied SHAP (SHapley Additive exPlanations) to quantify each input feature’s contribution to the final output, offering a global view of model behavior grounded in game theory.

These methods provide a multi-faceted understanding of how the model arrives at its predictions, promoting transparency and supporting analyst-driven interpretation in high-stakes financial contexts.

6.3 Large Language Models Reasoning

To construct a high-quality chain-of-thought dataset, we focused on identifying key terms in financial texts that align with sentiment labels generated by FinBERT. Our objective was to enhance the reasoning ability of large language models (LLMs) by incorporating signals from explainability methods. Specifically, we investigated whether integrating insights from attention weights, LIME, and SHAP could guide LLMs to outperform a standard zero-shot prompting baseline.

By embedding attention-guided keywords and interpretable features into the prompting process, we aimed to steer the model toward more accurate and transparent reasoning paths. This approach not only seeks to improve classification accuracy but also promotes more robust and explainable sentiment analysis, bridging the gap between predictive performance and interpretability in financial NLP.

7. Model Results & Analysis

7.1 Evaluation Metrics Used and Benchmarks

To evaluate our model performance, the evaluation metric used was standard classification metrics: accuracy, the proportion of correct predictions for the predicted label versus the ground truth label, precision, the proportion of correct positive sentiment predictions by the model, recall, the proportion of correct negative sentiment predictions by the model, and F1-Scores, which calculate the Harmonic mean of precision and recall. Our predicted labels were computed and compared against the ground truth for calculating the metrics. This evaluation framework allowed us to benchmark various approaches for finding the highest accuracy and improvement from the baseline of the BERT model sentiment predictions.

In addition to evaluating classification performance, we investigate whether incorporating explainability methods—such as the most attended tokens or important words identified by LIME and SHAP—can enhance the quality of reasoning generated by LLMs. Specifically, we compare two setups: (1) baseline reasoning generated directly from the input sentence, and (2) reasoning augmented with attention-based or explainability-informed insights. We employ an LLM-based evaluation framework to assess the generated reasoning across five dimensions: Helpfulness (how well the reasoning supports the sentiment label), Truthfulness (factual alignment with the sentence content and ground truth), Insightfulness (depth and originality of interpretation), Relevance (focus on sentiment-relevant parts of the sentence), and Clarity (coherence and readability of the explanation).

7.2 Discuss Results and Performance of Experiments

The accuracy of the baseline model using a fine-tuned BERT model is 88.59%. Our first experiment was trying to show if the highest weighted attention words outputted from the BERT model, along with a chain of thought reasoning, could prompt to guide the large language model to provide both reasoning and also increase the accuracy of results. All of the tests ran were using gpt-4o model from OpenAI.

	Value
Accuracy	0.89499
Precision (Positive)	0.89527
Recall (Positive)	0.892256
F1-Score (Positive)	0.893761
Precision (Negative)	0.874214
Recall (Negative)	0.874214
F1-Score (Negative)	0.874214
Precision (Neutral)	0.948052
Recall (Neutral)	0.949814
F1-Score (Neutral)	0.948932

Table 1. Evaluation Metrics using a fine-tuned BERT model

	Value
Accuracy	0.843137
Precision (Positive)	0.783172
Recall (Positive)	0.849123
F1-Score (Positive)	0.814815
Precision (Negative)	0.85625
Recall (Negative)	0.895425
F1-Score (Negative)	0.875399
Precision (Neutral)	0.877756
Recall (Neutral)	0.835878
F1-Score (Neutral)	0.856305

Table 2. Evaluation Metrics using 5 Attention weights along with few-shot prompting + chain of thought

Value		
	Accuracy	0.850361
Precision	(Positive)	0.805369
	Recall (Positive)	0.842105
	F1-Score (Positive)	0.823328
Precision	(Negative)	0.852761
	Recall (Negative)	0.908497
	F1-Score (Negative)	0.879747
Precision	(Neutral)	0.875984
	Recall (Neutral)	0.849237
	F1-Score (Neutral)	0.862403

Table 3. Evaluation Metrics using 3 Attention weights along with few-shot prompting + chain of thought

		Value
	Accuracy	0.885965
Precision	(Positive)	0.956679
	Recall (Positive)	0.892256
	F1-Score (Positive)	0.923345
Precision	(Negative)	0.92053
	Recall (Negative)	0.874214
	F1-Score (Negative)	0.896774

Table 4. Evaluation Metrics using a finetuned BERT model only for the positive and negative classes

		Value
	Accuracy	0.986273
Precision	(Positive)	0.998507
	Recall (Positive)	0.981658
	F1-Score (Positive)	0.990011
Precision	(Negative)	0.967846
	Recall (Negative)	0.996688
	F1-Score (Negative)	0.982055

Table 5. Evaluation Metrics using 3 Attention weights along with few-shot prompting + chain of thought only for the positive and negative classes

		Value
	Accuracy	0.990849
Precision	(Positive)	0.997779
	Recall (Positive)	0.988995

F1-Score (Positive)	0.993368
Precision (Negative)	0.977236
Recall (Negative)	0.995033
F1-Score (Negative)	0.986054

Table 6. Evaluation Metrics using only zero-shot prompting thought for the positive and negative classes

	Helpfulness	Truthfulness	Insightfulness	Relevance	Clarity
Zero-shot Reasoning	4.67	4.82	3.84	4.84	4.95
Augmented with Attention Reasoning	4.86	4.81	4.01	4.87	4.94

Table 7. Evaluation Metrics for zero-shot reasoning and augmented reasoning

7.3 Analysis of the Results of Our Work

Text	Ground Truth	Predicted Label	Attention Reasoning	Zero-Shot Prediction	Zero-Shot Reasoning
Investors will continue being interested in the company 's share although it is not quite cheap , Affarsvarlden said .	positive	negative	The word 'not' combined with 'cheap' suggests a negative sentiment as it implies the shares are expensive, which is generally unfavorable for investors.	positive	The statement suggests that despite the company's shares not being cheap, there is ongoing interest from investors, indicating optimism and a favorable outlook towards the company's potential or performance.

Fig. 3. Failure case of zero-shot prompting outdid attention word-based reasoning

In our experiments, we found that using attention weights as guidance for sentiment predictions for large language models resulted in significantly worse performance on the multiclass sentiment classification task. From the analysis of failure cases, using the attention weights as guides for the large language model to predict sentiment outcomes had negative effects for the neutral class, as many positive cases were labeled into neutral as well as negative cases.

Therefore, to further explore to see if attention weights from BERT models can become useful guides for reasoning performance improvements, we decided to remove the neutral class and turn

the classification to binary (positive and negative). By doing so, significant improvements were achieved using a large language model compared to the baseline BERT model in both cases, using attention weights and zero-shot prompting.

Our results found significant improvements with a ~11% improvement using attention weights as a reasoning guide. However, attention weights as a reasoning guide still fell short of zero-shot prompting (no attention weights as guides) with a 0.04% difference in overall accuracy. These findings suggest that while attention weights may worsen performance in multiclass settings, they can still provide meaningful guidance in binary classification, though they may not always improve the effectiveness of pure zero-shot prompting.

Furthermore, we evaluated the reasoning quality produced by models using zero-shot prompting and attention-augmented reasoning. Augmenting reasoning with attention-guided words resulted in consistent improvements in Helpfulness (+0.19) and Insightfulness (+0.17), while maintaining comparable scores in Truthfulness, Clarity, and Relevance. These results indicate that although attention signals may introduce noise into label predictions, they can enhance the interpretability and depth of the model's explanations. Thus, attention-based reasoning can act as a valuable complementary signal for generating richer, more informative rationales—even when it does not consistently improve classification accuracy.

8. Correlation Between Market Sentiment and Financial Metrics

8.1 Methodology of Aggregating and Normalizing

We organize the collected monthly or daily financial news into a standardized format, ensuring that each news text is limited to a specific number of words, such as 100 words, for efficient model processing. The text is then input into a BERT NLP model, which generates a CLS representative vector, and a classifier maps each text segment to its corresponding sentiment label and sentiment score.

We collect representative financial metrics from publicly available sources, such as P/E ratio, Return on Equity (ROE), EBITDA, Net Income, etc., compile them into a file, and arrange them on a quarterly timeline. Since sentiment scores range from -1 to 1, while most financial metrics range from -1000 to 100,000 million, we normalize these large numerical values to a comparable range to improve the visualization of time series graphs and observe trends more effectively.

Additionally, we calculate the Pearson Correlation Coefficient between sentiment scores and various financial metrics to analyze how different metrics impact market sentiment. Since financial metrics follow a quarterly timeline, whereas news sentiment scores follow a daily or monthly timeline, we average sentiment scores on a quarterly basis, where the average sentiment score from January to March is matched with Q1 financial metrics.

8.2 Financial Correlation Objective and Insights

Our goal is to identify the most explainable financial metrics that influence market sentiment, helping investors better track market trends and make data-driven decisions with the support of cutting-edge AI.

Based on the existing results, we found that for companies in different industries, the influence and explainability of financial metrics on market sentiment vary significantly. For example, in the financial services industry, such as Visa, the P/E ratio shows a strong negative correlation, meaning that when P/E decreases, market sentiment rises, and vice versa. A declining P/E ratio indicates a downward trend in the company's stock price, which is seen as a positive signal for many financial industry investors, as they often seek to buy at the lowest price and sell at the highest price to profit from arbitrage.

Next, we aim to collect a reasonable amount of data from companies across different industries and identify the financial metrics with the highest explanatory power for market sentiment in each industry.

8.3 FinTech Industry Specific Preparation

Each industry is represented by five influential companies, and the most influential and interpretable financial metrics on market sentiment are identified for each sector. We first focus on the Financial Technology Industry.

First, by using APIs, searchable AI tools, and visiting company websites, we collect the most representative monthly financial news articles from companies within the industry. The integrated format includes: publication date, news title, summary of the content, link, and sentiment score. To ensure the quality and accuracy of the news, we manually click each link to verify that the news actually exists and that the date and content are correct.

Second, we input the collected financial texts into a BERT model to generate sentiment scores for each article. Third, we extract each company's quarterly financial metrics from Macrotrends, including Net Income, EBITDA, Total Debt, Total Assets, Revenue, and ROI. Finally, we integrate and normalize the sentiment scores and financial metric data, and use Pearson Correlation Coefficients, regression tables, time series graphs, and other visualizations to examine the correlation and interpretability between financial metrics and market sentiment.

8.4 FinTech Industry Outcome and Analysis

The five FinTech companies we selected are PayPal, Fiserv, Intuit, Block, and JPMorgan, and we calculated the correlation coefficients for each. Based on the results as shown in Figure 4, EBITDA generally shows a positive correlation across all companies. This suggests that a higher

EBITDA tends to have a positive impact on market sentiment in the FinTech industry. EBITDA represents a company's earnings in an ideal state—excluding taxes and interest—thus eliminating macro-level uncertainties such as political or economic factors. It better reflects the operational strength of a company. A higher EBITDA indicates stronger performance during the quarter, which tends to boost market sentiment, explaining its positive correlation.

As for ROI, its influence on overall market sentiment also tends to be positively correlated. A higher return on investment signals that more investors are willing to enter the market, thus driving positive sentiment. In contrast, indicators like Net Income, Revenue, and Total Assets exhibit larger error margins, meaning they are influenced by numerous uncontrollable market factors, leading to some degree of randomness and uncertainty in their correlation with sentiment.

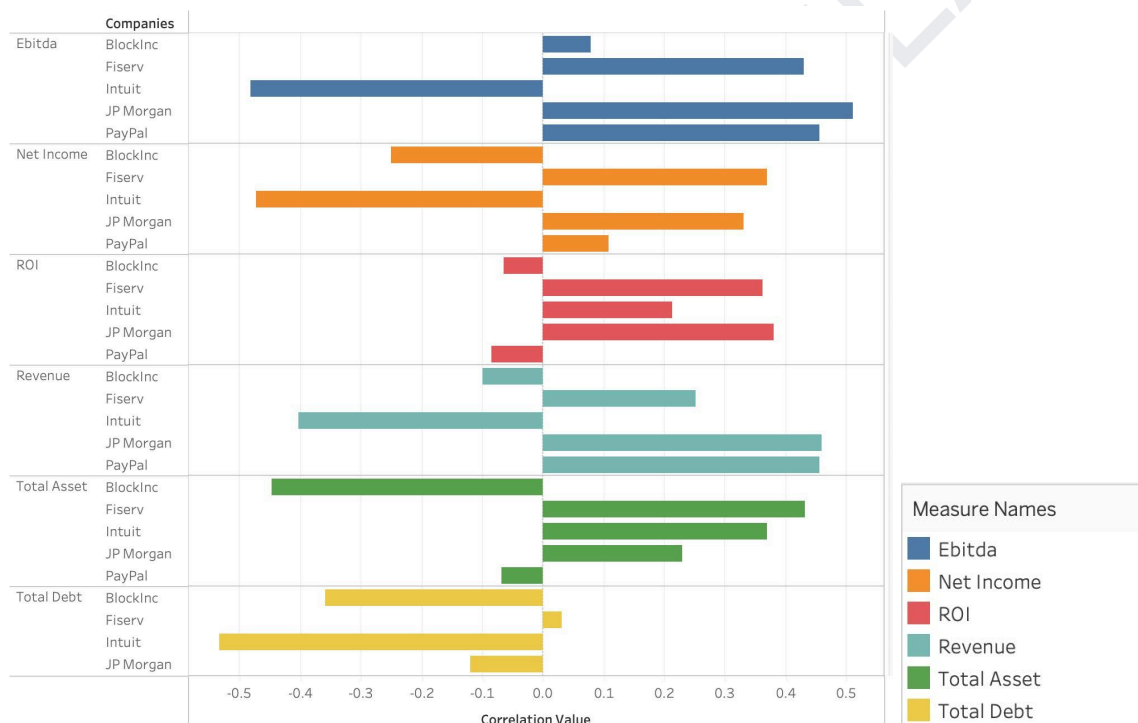


Fig. 4. Correlation Across FinTech Industry

Another relatively interpretable indicator is Total Debt, which shows a negative correlation with overall market sentiment. This is because the level of corporate debt reflects the level of investment risk—generally, FinTech companies with higher debt levels are seen as riskier, leading to a negative impact on sentiment. To better visualize the impact of each metric within the industry, we took the average values across the selected companies, as illustrated in Figure 5. The metrics with a stronger influence on FinTech market sentiment include EBITDA and ROI, which show positive correlations, and Total Debt, which shows a negative correlation.

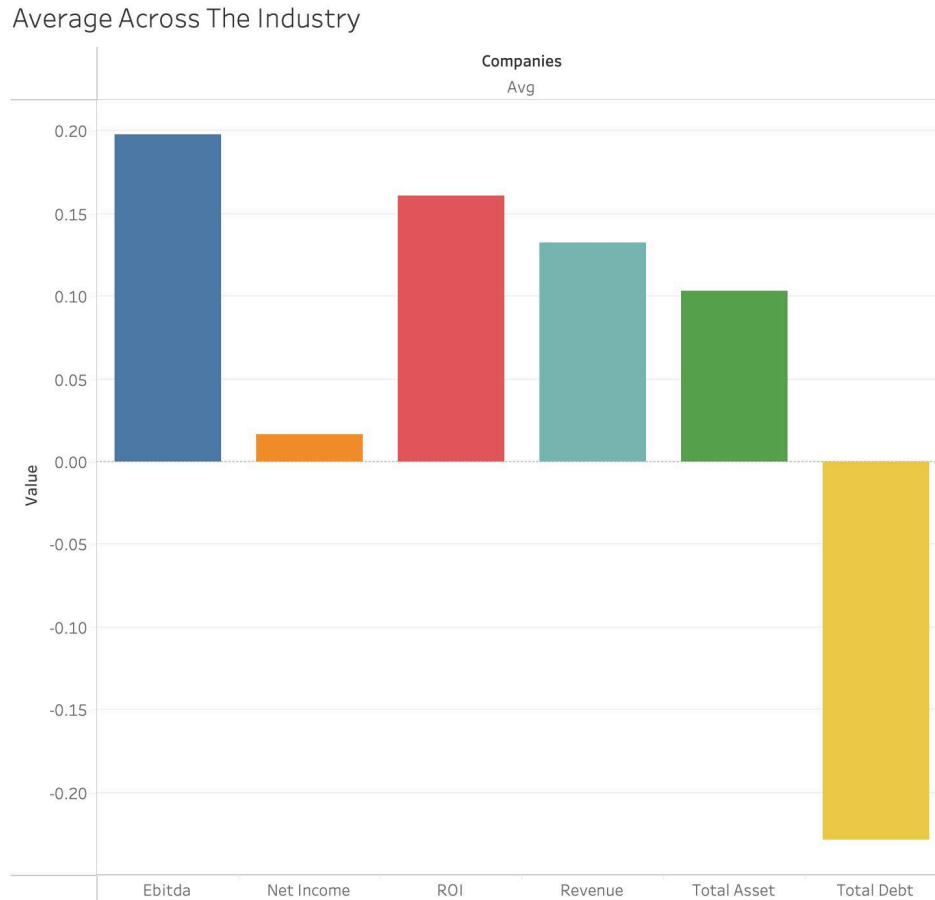


Fig. 5. Averaged Correlation Across FinTech Industry

By examining highly correlated metrics and market sentiment through a time series graph, using PayPal's EBITDA as an example (see Figure 6), we can observe that their overall trends, even in subtle increases, are very closely aligned. However, the sentiment trend is more volatile because market sentiment is typically influenced in real-time by daily news or events, while EBITDA reflects the company's gradual financial trend and is therefore smoother.

PayP

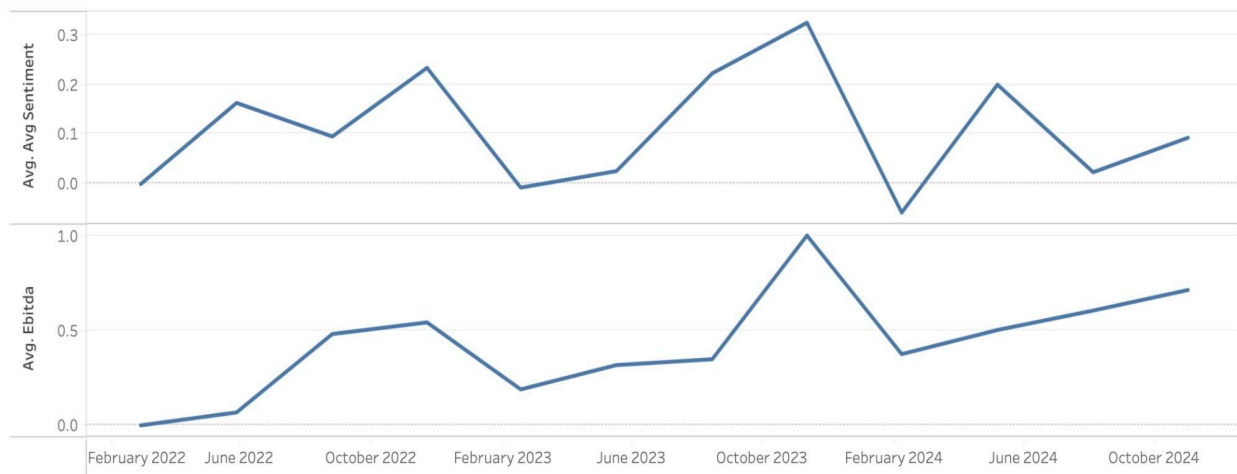


Fig. 6. Time Series EBITDA & Sentiment

8.5 Consumer Staples Industry Outcome and Analysis

Apart from the fintech industry, we also took a big dive into the consumer staples industry. The seven companies targeted in our analysis—Coca-Cola, Hershey, Mondelez, Monster Beverage, Philip Morris, Procter and Gamble, and Anheuser-Busch InBev—are an assorted group of beverage, snack, tobacco, and personal care products. We calculated the Pearson correlation coefficients of quarterly news sentiment vs. the top five financial metrics: Net Income, Quarterly Revenue, EBITDA, Total Assets, and Long-Term Debt. The analysis below provides insight into how sentiment tracks with financial results in an industry as much driven by consumer tastes as the consumer staples group is.

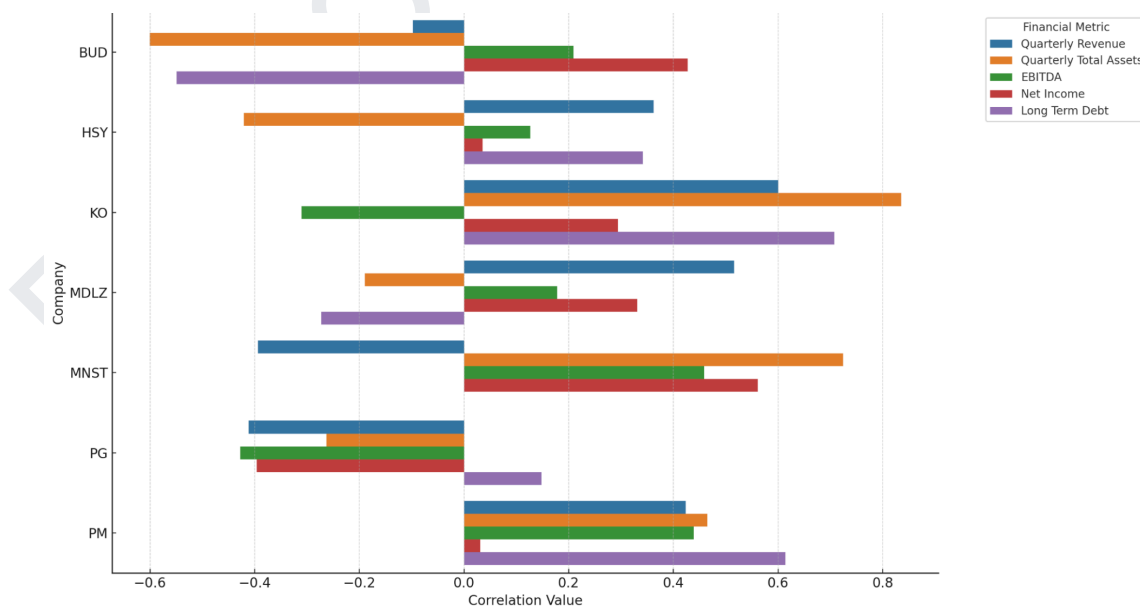


Fig. 7. Correlation Across Consumer Staples Industry

Net Income which illustrated below in Figure 8 proved to be the most persistently correlated metric with news sentiment at most companies. Firms like Monster and Coca-Cola demonstrated good ties between bullish media tone and improved quarterly earnings. This trend backed up the idea that consumer brands are positively impacted when good stories drive demand, confidence, or price resilience. The relationship between sentiment and net income performance indicates that public tone tends to reflect real-time profit momentum.

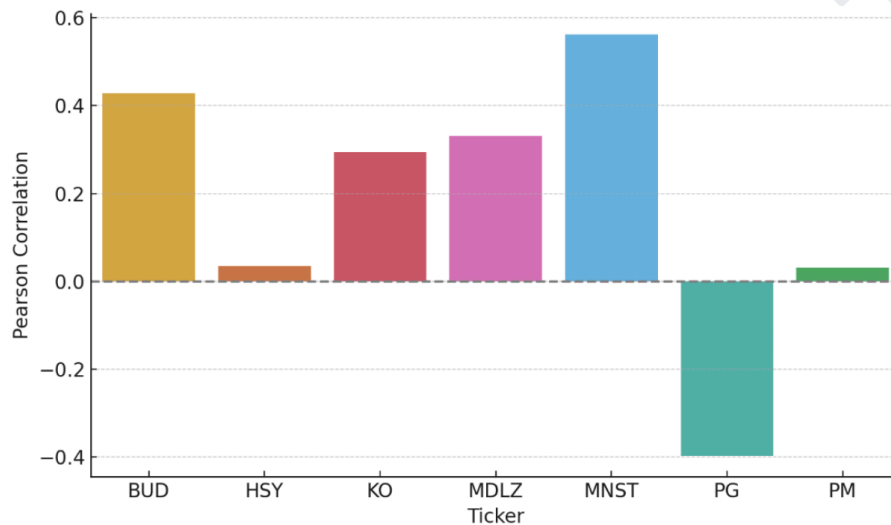


Fig. 8. Correlation between Net Income and Sentiment in Consumer Staples Industry

Total Assets, however, had mixed trends. For the likes of Coca-Cola and Monster, high and positively correlated growth indicates that growth in assets follows or occurs along with good sentiment. This is probably because of investing in manufacturing capacity, product growth, or higher inventory in the face of increasing demand. But the likes of Procter and Gamble and Anheuser-Busch had weak or negative trends, which can imply that during good sentiment, these companies are decreasing the asset base or engaged in capital discipline. The opposite results indicate different strategic stances where some companies take the good sentiment period to invest and the others are concerned with operational efficiencies or selling off businesses.

Both EBITDA, Quarterly Revenue, Long-Term Debt yielded weak and overall moderate correlations. Although there were weak to moderate positive associations across many of the companies, the overall absence of statistical significance indicates that these operating metrics are driven by a range of factors beyond the scope of media tone. Cost structure volatility, the timing of promotions, or pricing strategy might mask the relationship between sentiment and

underlying operating results. These results imply sentiment can indicate high-level financial momentum but does not necessarily reflect the nuances of short-term revenue or margin results.

Collectively, the strongest gauge of the connection between public tone and financial wellness in consumer staples is net income. The relationship between sentiment and balance sheet size, leverage, or underlying business seems to vary on a case-by-case basis depending on strategy and position. Positively oriented sentiment is not always an unequivocal indicator but, in the context of the company, is meaningful in the direction of future financials.

8.6 Recommendations for Future Correlation Analysis

We believe that for different industries, financial metrics with high correlation to market sentiment can be identified, which in turn can help private investors make data-driven decisions. In addition to industry-based segmentation, other characteristics could also be used for classification—for example, distinguishing between high-capital and low-capital companies, or between companies engaged in international cooperation and those that are domestic-focused. Next, we may conduct deeper research on this and also explore the most interpretable financial metrics in the Pharmacy and Customer Service industries.

9. Limitations & Future Work

9.1 Create a High-Quality Chain of Thought Reasoning Dataset

The current motivation of our reasoning outputs is to determine the method that has the highest accuracy in sentiment and finding what contributes to driving a sentiment to the respective positive, neutral, or negative class. Having this understanding in the financial context will provide even better reasoning datasets. However, after running experiments, we see that words with the highest attention weights are not accurate predictors of sentiment class. Therefore, future work would be to continuously refine this chain of thought reasoning dataset by seeing if, given the ground truth of the sentiment along with the highest attention weight words that the model can generate accurate and high-quality reasoning.

9.2 Train Reasoning Models Specific to Explainable Sentiment Analysis

Currently, the explainable sentiment from the chain of thought prompting is not of high quality to further train downstream applications. To address these limitations, the goal should be crafting a dataset that aligns with human expectations through reinforcement learning from human feedback to teach models to align models' reasoning outputs more closely to how humans reason and interpret sentiment. A high-quality reasoning dataset can train reasoning models (or

transformer-based models) specific to explainable sentiment analysis and create models that can benefit stakeholders in the financial industry.

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